

CSCE 676 DATA MINING · MODULE 08 · LECTURE NOTES

Anomaly Detection

Unusual compared to what?

MODULE OUTCOMES

Explain the three problem settings and why they differ.

Explain the method families and their assumptions.

Explain the base rate problem in this setting.

Explain why evaluation is especially hard here.

Deploy a detector with a defensible threshold.

1 Three problems

Setting	What you have	What follows
Unsupervised	Unlabelled data, assumed to be mostly normal.	You find statistical outliers, which are not necessarily the anomalies you care about — see the callout.
Semi-supervised (one-class)	A clean sample of normal data, and nothing else.	You model normality and flag departures from it, which is a better-posed problem than the first row.
Supervised	Labelled examples of both normal and anomalous records.	This is imbalanced classification (CSCE 633) — see the note.

The third row is the one to recognise early: if you have labels for both classes, this is a classification problem with a class imbalance, and the anomaly detection literature is the wrong one to read. Use a classifier, handle the imbalance with the standard methods, and evaluate with precision and recall. **Identifying this case saves people a great deal of wasted effort**, because the unsupervised methods will underperform a straightforward classifier that uses the labels you already have.

STATISTICALLY UNUSUAL AND OPERATIONALLY INTERESTING ARE DIFFERENT PROPERTIES

An unsupervised detector finds records far from the rest of the distribution — which includes measurement errors, missing-value sentinels (the row where age is 999), duplicated rows, encoding failures, and genuinely legitimate rare events, as well as whatever it was you were hoping to find.

And the ratio is unfavourable. In most real data, the top outliers are overwhelmingly data quality problems — which is a genuinely useful finding and is not the one you wanted, and discovering it is a rite of passage with these methods.

So the first run of any detector is a data cleaning exercise, and expecting otherwise is the commonest disappointment with this material. Budget for it, and treat the cleaning as a deliverable.

Which means you have to specify what kind of unusual you care about — a feature set that excludes the artefacts, or a semi-supervised setup with verified clean normal data — both of which require knowing the domain, and neither of which is a modelling decision you can make from the data alone.

2 The methods

STATISTICAL

fit a distribution, flag the low-probability tail. Assumes you know the distribution. Robust versions use the median and the median absolute deviation rather than the mean and standard deviation, because the outliers corrupt those estimates -- which is the whole point of robust statistics.

DISTANCE AND DENSITY

kNN distance, or Local Outlier Factor, which compares a point's local density to its neighbours'. LOF handles varying density; both suffer from Module 05's distance concentration in high dimension.

ISOLATION-BASED

Isolation Forest: random splits isolate outliers in fewer splits than inliers. Fast, near-linear, few assumptions, and the right default for tabular data.

RECONSTRUCTION

autoencoder or PCA residual: flag whatever the model cannot reconstruct well. Assumes normal data lies on a learnable manifold.

AND EACH DEFINES "normal" DIFFERENTLY, so they disagree -- which is Module 04's situation again.

Isolation Forest is the right default for tabular data — few assumptions, near-linear time, and it degrades gracefully in high dimension where the distance-based methods degrade badly (Module 05 §1).

Giving a default is worth doing, because the alternative is trying all four families and then having no principled way to choose between their disagreeing outputs — which is Module 04 §3's circularity in a new setting.

3 The base rate

$$\text{precision} = \text{TP} / (\text{TP} + \text{FP})$$

- Precision is the fraction of flagged records that are genuinely anomalous — and it is the number a human reviewer actually experiences, which is why it governs whether a system is usable.
- With a base rate of 0.1%, a recall of 90%, and a false positive rate of 1%, the precision is about 8% — so 92 of every 100 alerts are false, from a detector whose stated performance sounds excellent.
- And at a base rate of 0.01%, the precision falls to about 0.9% — a tenfold rarer event makes the same detector unusable, with no change to the detector at all. The detector's quality did not move; the deployment became impossible.
- Which is why a 99%-accurate detector produces mostly false positives on rare events — and this is CSCE 701 Module 09 §3's arithmetic, in its original setting, where it was first understood and is still most often ignored.

SO THE DESIGN QUESTION IS REVIEW CAPACITY, NOT DETECTOR ACCURACY

Work backwards from how many records a person can realistically review per day — and set the threshold so that the alert volume matches that capacity, then report the recall you achieved at that volume.

Which inverts the usual framing entirely. 'What recall can we achieve at fifty alerts per day' is answerable and actionable; 'what is our AUC' is neither, because it does not correspond to any deployment you would choose.

And correlate weak signals rather than trying to improve a single detector — three independent weak signals on the same record beats any one of them substantially, because the base rate improves multiplicatively while the recall falls only somewhat (CSCE 701 Module 09 §3).

Or change the problem altogether: rank rather than classify, and present a reviewer with the top k ranked records with the reasons attached — which is a usable system in circumstances where a classifier is not, and is how most successful deployments of this material actually work.

4 Evaluation

- You have almost no labels, and the few you have are the anomalies somebody already found — which biases the evaluation set toward the easy ones and systematically overstates performance on the hard cases that matter.
- Accuracy is useless here. Predicting 'normal' for everything gives 99.9% accuracy at a 0.1% base rate, so report precision and recall at a specific threshold and never an accuracy figure.
- AUC flatters. It averages over thresholds you would never actually use — including those producing thousands of alerts — so prefer precision at the k you can genuinely review, or average precision, which weights the useful region.

- **And injected synthetic anomalies are systematically easier than real ones**, because you generated them from your own idea of what anomalous looks like — **so a detector evaluated on them is overstated**, frequently by a large factor.
- **What works: precision at k established by expert review of the top k, plus the time to detection on historical incidents you know about. Time to detection on known past incidents is the most convincing evaluation available** — it uses real anomalies, requires no new labelling, and measures the thing the system is actually for.

AND THE HONEST SUMMARY OF THIS MATERIAL

Anomaly detection is reliably oversold, because **the methods are easy to run and the base rate makes them hard to deploy** — and **the gap between those two facts is invisible until somebody actually reviews the alerts**, which is usually after the system has been bought.

It works best where the base rate is not tiny — manufacturing quality control, systems monitoring, and data cleaning — **and worst where it is**, which unfortunately includes most fraud and security framings, where it is most often proposed.

And it works best as a ranking presented to an expert rather than as a decision (§3) — which is a smaller claim about the technology and a more defensible system.

So the claim to make is narrow: 'at fifty alerts per day, the detector surfaces 30% of known incidents a median of four hours earlier than they were otherwise found' — **which is checkable, modest, and genuinely useful**, and is exactly the form Module 13 asks for.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Aggarwal — Outlier Analysis, 2nd edition	The whole module — and its chapter on evaluation is the most honest treatment available. Library copy.
Liu, Ting & Zhou — Isolation Forest (free)	§2's default method in the original — short, and the intuition is well explained.
Breunig et al. — LOF: Identifying Density-Based Local Outliers (free)	§2's density row — and the varying-density problem it was designed to solve.
Campos et al. — On the evaluation of unsupervised outlier detection (free)	§4's problem, measured — a systematic study showing how much the evaluation choices change the conclusions.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Classify your own problem** into one of the three settings, and say what follows.
- 2 **Run a detector on a real dataset** and read the top fifty outliers.
- 3 **Categorise them** as data quality, legitimate rare events, or interesting.
- 4 **Run all four method families** and measure how much their top fifty overlap.
- 5 **Compute precision** at base rates of 1%, 0.1%, and 0.01% with fixed recall and FPR.
- 6 **Set a threshold from a stated review capacity** and report the recall you get.
- 7 **Correlate three weak detectors** and measure the precision improvement.
- 8 **Compute accuracy, AUC, and precision at k** for one detector, and say which you would report.
- 9 **Inject synthetic anomalies and compare** the detector's performance against its performance on real ones.
- 10 **Measure time to detection** on an incident you know the date of.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the three settings and what follows from each.
- 2 Which should you recognise immediately, and why?
- 3 Why are statistically unusual and operationally interesting different?
- 4 What are the top outliers in real data usually?
- 5 Name four method families and what each assumes.
- 6 Which is the right default, and why?
- 7 Give the precision arithmetic at a 0.1% base rate.
- 8 What is the right design question, and how does it invert the usual framing?
- 9 Give four reasons evaluation is hard here.
- 10 What is the most convincing evaluation available?